

A Non-Positive Kazhdan Projection for a Banach-Space Representation Family

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Status

This packet gives a candidate counterexample to the general question in Section 3.5 of de la Salle [1]: if $\mathcal{C}_{\mathcal{F}}(G)$ contains a Kazhdan projection, must that projection belong to the closure of the nonnegative compactly supported functions of integral one?

The answer is no for arbitrary Banach-space representation families. The example is finite-dimensional and non-isometric. It does not contradict the source paper's positive result for central Kazhdan projections attached to isometric representation families.

Source Question

In Section 3.5, after defining the positive variant of a Kazhdan projection, the source paper states:

3.5. Positive Kazhdan projections. A natural variant of Definition 3.1 is to require additionally that p belongs to the closure of the *nonnegative* functions $\{f \in C_c(G), f \geq 0, \int f = 1\}$. This variant is particularly relevant to the applications to dynamics, see [6]. To our knowledge, in all examples where a Kazhdan projection is known to exist (for examples strong property (T), or Corollaries 5.5 -5.11), it belongs to the closure of the nonnegative functions. However, we do not know if this is the case in general.

We can note that the proof of Theorem 1.2 shows in full generality that $\mathcal{C}_{\mathcal{F}}(G)$ contains such a “positive” Kazhdan projection if and only if there is a *positive*

The question is whether, in general, the existence of a Kazhdan projection forces the existence of one in the closure of nonnegative compactly supported probability functions. For the discrete group used below, these functions are exactly finitely supported probability measures on $\mathbb{Z}$.

Definitions Used

Let G be a locally compact group and let $\mathcal{F}$ be a set of Banach-space representations satisfying the uniform generator-bound hypothesis of the source. The seminorm on $C_c(G)$ is

$$\|a\|_{\mathcal{F}} = \sup_{(\pi, E) \in \mathcal{F}} \|\pi(a)\|_{B(E)},$$

and $\mathcal{C}_{\mathcal{F}}(G)$ is the corresponding completion.

A Kazhdan projection is an idempotent $p \in \mathcal{C}_{\mathcal{F}}(G)$, lying in the closure of functions with integral one, such that $\pi(p)$ is the projection onto E^π for every $(\pi, E) \in \mathcal{F}$. The positive variant asks that p lie in the closure of the nonnegative functions with integral one.

Proof Intuition

The ordinary Kazhdan projection only asks for approximation by signed mass-one functions. A signed Laurent polynomial can annihilate two spectral values on opposite sides of the unit circle at the same time. Positive probability measures cannot do this: if N is an integer-valued random variable, the two moments $E(2^N)$ and $E(2^{-N})$ cannot both be small because $2^N + 2^{-N} \geq 2$ pointwise. This separates signed Kazhdan projections from positive ones in the smallest possible diagonal example.

Theorem 1. *There exist a compactly generated infinite group G , a family $\mathcal{F}$ of Banach-space representations satisfying the source hypotheses, and a Kazhdan projection in $\mathcal{C}_{\mathcal{F}}(G)$ which is not in the closure of the nonnegative compactly supported probability functions.*

Proof. Let $G = \mathbb{Z}$, written multiplicatively with generator u . Let $E = \mathbb{C}^3$ with the supremum norm, and define a representation $\pi: \mathbb{Z} \rightarrow GL(E)$ by

$$\pi(u) = T = \text{diag}(1, 2, 1/2).$$

Let $\mathcal{F} = \{(\pi, E)\}$. If $S = \{-1, 0, 1\}$, then $\sup_{s \in S} \|\pi(s)\| = 2$, so the source's generator-bound hypothesis is satisfied.

For a finitely supported function $a = \sum_n a_n u^n \in \mathbb{C}[\mathbb{Z}]$, write

$$\hat{a}(\lambda) = \sum_n a_n \lambda^n.$$

Then

$$\pi(a) = \text{diag}(\hat{a}(1), \hat{a}(2), \hat{a}(1/2)).$$

The invariant vectors of π are exactly

$$E^\pi = \mathbb{C}e_1.$$

Consider the signed mass-one group-algebra element

$$p = -2u^2 + 5u - 2.$$

Its total mass is $p(1) = -2 + 5 - 2 = 1$, and

$$p(1) = 1, \quad p(2) = 0, \quad p(1/2) = 0.$$

Hence

$$\pi(p) = \text{diag}(1, 0, 0),$$

the projection onto E^π . Also $\pi(p)^2 = \pi(p)$, so p is idempotent in the quotient completion $\mathcal{C}_{\mathcal{F}}(\mathbb{Z})$. Thus p is a Kazhdan projection in the sense of the source paper.

It remains to prove that no positive probability approximation exists. Let μ be a finitely supported probability measure on $\mathbb{Z}$. Then

$$\pi(\mu) = \text{diag}\left(1, \sum_n \mu(n)2^n, \sum_n \mu(n)2^{-n}\right).$$

If a net μ_i of such measures converged to p in $\mathcal{C}_{\mathcal{F}}(\mathbb{Z})$, then both

$$A_i = \sum_n \mu_i(n) 2^n, \quad B_i = \sum_n \mu_i(n) 2^{-n}$$

would tend to zero. This is impossible, because for every i ,

$$A_i + B_i = \sum_n \mu_i(n) (2^n + 2^{-n}) \geq 2 \sum_n \mu_i(n) = 2.$$

Therefore p is not in the closure of the nonnegative probability measures. □

Verification Notes

The construction uses the infinite compactly generated group $\mathbb{Z}$, so it fits the source’s standing locally compact compactly generated setting. The representation is strongly continuous because G is discrete. The family $\mathcal{F}$ has one representation, and the generator norms are finite. The completion is finite-dimensional: it is the image of $\mathbb{C}[\mathbb{Z}]$ under

$$a \mapsto (\hat{a}(1), \hat{a}(2), \hat{a}(1/2)).$$

Thus all convergence statements reduce to convergence of the two scalar moments at 2 and 1/2.

Limitations

The counterexample is literal for arbitrary Banach-space representation families. It is not an example from strong property (T), robust property (T), or isometric representation theory. It also does not refute Proposition 3.13 of the source, which assumes an isometric family stable by complex conjugation.

Novelty Check

The run indexes were searched for arXiv:1604.01616, the title, Kazhdan projection, positive Kazhdan projection, nonnegative functions, Banach property (T), fixed point spectrum, Laplacian, cohomology, and expander keywords. No exact packet for this source question was found. Nearby indexed results concern BFGM higher-rank fixed point properties, fixed point spectra, and expander/superexpander questions.

External phrase searches on June 28, 2026 used the phrases **positive Kazhdan projections**, **nonnegative functions**, **Kazhdan projection**, **closure of the nonnegative functions**, and the exact source sentence. They did not locate a published answer. The novelty confidence is therefore bounded by phrase search, but the example is self-contained.

Human Review Recommendation

Human review should check the interpretation of the phrase “in general” in Section 3.5. If it is read in the full generality of the paper’s arbitrary Banach-space representation families, the counterexample is complete. If the intended question was restricted to isometric, self-adjoint, or naturally arising strong-property-(T) families, this packet should be reclassified as a scope caveat rather than a resolution of that narrower problem.

References

- [1] M. de la Salle, *A local characterization of Kazhdan projections and applications*, arXiv:1604.01616.